MIL-F-3192
28 APRIL 1950
SUPERSEDING
Navy 53F9(Ord)
6 July 1948

MILITARY SPECIFICATION

FIBERBOARD, FLAME-RESISTANT (FOR PROJECTILE GROMMETS)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments

1. SCOPE

1.1 The flame-resistant fiberboard covered by this specification is used in the manufacture of fiber grommets for the protection of rotating bands of projectiles during handling, shipment, and storage.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 **Specifications.**—The following specifications of the issue in effect on date of invitation for bids, form a part of this specification.

U S ARMY SPECIFICATION

100-2—Standard Specification for Marking Shipments by Contractors

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[1]

(*Army*—Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Navy*—Copies of Federal, Military, and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D C, except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Center, Norfolk 11, Va.)

2.2 **Other publications.**—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

[1] Applicable only to Navy purchases

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook

(Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D C, except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Center, Norfolk 11, Va.)

3. REQUIREMENTS

3.1 **Material.**—The fiberboard shall be made from cotton, hemp, flax or wood fibers, or a combination thereof so treated as to be water and flame resistant.

3.2 **Physical and chemical properties.**—The fiberboard shall comply with table I.

TABLE I—*Physical and chemical properties*

Property	Required value
Bursting strength, p s i per mil thickness	6 (min.)
Tensile strength, p s i	
a. Machine direction	5,000 (min.)
b. Cross-machine direction	1,800 (min.)
Compression set, percent.	
a. As received (conditioned)	25 (max.)
b. After baking for 24 hours at 212° F.	25 (max.)
c. After immersion for 24 hours in water at 70° to 75° F	45 (max.)
Water absorption in 24 hours at 70° to 75° F., percent	50 (max.)
Burning rate, inches per minute	
a. As received (conditioned)	1 (max.).
b. After immersion in water and subsequent drying in air	1 (max.)
Chlorides, calculated as zinc chloride, percent.	0.10 (max.)
Ash, percent	30 (max.).

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3.3 **Flexibility.**—There shall be no cracks or separation of the laminations of fiberboard when tested as specified in 4.3.10.

3.4 **Delamination.**—There shall be no delamination of fiberboard when tested as specified in 4.3.11.

3.5 **Finish.**—The fiberboard shall have a smooth finish.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**

4.1.1 *Lot.*—A lot shall consist of one mill run of fiberboard.

4.1.2 *Sampling for lot tests.*—The Government inspector shall select samples consisting of at least 2 square feet of fiberboard from the beginning, middle and end of the mill run. The samples shall be marked to indicate machine and cross-machine directions. On request of the Government inspector the samples shall be [illegible] laboratory marked as follows:

Samples for lot acceptance
Lot number ____ Contract number ____
Fiberboard, Flame Resistant
Specification MIL-F-3192
Contractor's name ____
Submitted by ____ Date ____

4.1.3 *Check tests.*—The Government inspector may, from time to time, forward additional samples of the fiberboard to a Government laboratory for any or all tests.

4.2 **Inspection.**

4.2.1 Unless otherwise specified, inspection shall be at the point of manufacture.

4.2.2 The fiberboard shall be inspected by the Government inspector to determine compliance with the requirements of this specification that do not involve test.

4.3 **Tests.**—Tests shall be conducted at a Government laboratory in accordance with the following.

4.3.1 *Number of specimens.*—Three specimens of every sample taken as specified in 4.1.2 shall be tested for each requirement.

4.3.2 *Standard testing conditions.*—Samples shall be tested immediately after conditioning for at least 72 hours under uniform atmospheric conditions of 70° to 75° F. temperature and 50 percent relative humidity. In cases of dispute, all physical tests shall be conducted in an atmosphere so conditioned.

4.3.3 *Bursting strength.*

4.3.3.1 *Apparatus.*—The apparatus employed shall be of the Jumbo Mullen or Cady type having concentric circular apertures 1.24 inches in diameter.

4.3.3.2 *Procedure.*—A specimen shall be clamped firmly in the machine so as to prevent slipping. Pressure shall be applied by action of a standard motor drive or a handwheel. If a hand-driven testing machine is employed, the wheel shall be turned at a uniform speed of approximately two revolutions per second. If a specimen slips during a test, the result shall be disregarded. The indicated pressure in p. s. i. when the specimen breaks is the bursting strength.

4.3.4 *Tensile strength.*

4.3.4.1 *Test specimen.*—The test specimen [illegible] long.

4.3.4.2 *Procedure.*—The width and thickness of the specimen shall be measured before it is placed in the testing machine. The distance between clamps shall be not less than 6 inches. The loading speed shall be 12 inches per minute. Test results shall be disregarded if the specimen breaks within ½ inch of the jaws. The tensile strength shall be calculated in p. s. i.

4.3.5 *Compression set.*

4.3.5.1 *Apparatus.*—The apparatus shall consist of a hardened steel block, a hardened steel mandrel 1.129±0.0005 inches in diameter, and a suitable compression device.

4.3.5.2 *Procedure.*—Test specimens shall be cut approximately 2 inches square and the thickness measured to the nearest 0.001 inch at or near the center. The specimen shall be placed on the block in the compression device and a load of 5,000 pounds applied by means of the 1.129 inch diameter mandrel. The load shall be applied slowly and held for 30 seconds. Within 1 minute after removal of the load the thickness of the specimen shall be measured to the nearest 0.001 inch at or near the center of the impression and the percent compression set calculated. Specimens as follows shall be tested:

a. As received (after conditioning).

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b. After baking for 24 hours at 212° ±2° F.

c After immersion for 24 hours in water at 70° to 75° F.

4.3.6 *Water absorption*—Specimens shall be weighed and allowed to stand for 24 hours in a container of distilled water at 70° to 75° F. The specimens shall rest on edge and be completely submerged After 24 hours the specimens shall be removed from the water and wiped with a cloth or paper towel The specimen shall be weighed and the percent increase in weight calculated.

4.3.7 *Burning rate*

4.3.7.1 *Test specimen*—The test specimen shall be 6 ± 1/64 inches long and 1 ± 1/64 inches wide Lines spaced ¼ inch apart shall be scribed on the specimen in the direction of its width for a distance of 1½ to 2 inches from one end

4.3.7.2 *Apparatus*—The apparatus shall consist of a laboratory ring stand provided with a ring and one extra clamp, a Bunsen burner, a 5-inch square piece of 20-mesh gauze, and a stop-watch. (See fig 1)

4.3.7.3 *Procedure*—The specimen shall be clamped to the ring stand at the unmarked end with the longitudinal axis of the specimen horizontal and the transverse axis inclined 45 degrees to the horizontal, as shown on figure 1. The ring shall be clamped to the ring stand below the specimen, the wire gauze placed thereon and the ring adjusted so that the wire gauze is ¼ inch below the specimen with approximately ½ inch of the specimen extending beyond the wire gauze The tip of a Bunsen burner flame, adjusted to a length of ½ to ¾ inches shall be brought into contact with the free end of the specimen, held for 20 seconds, and then the burner shall be withdrawn. The rate at which the specimen burns along the bottom edge shall be recorded as the burning rate

4.3.8 *Chlorides calculated as zinc chloride*—Ten gm of the finely divided fiberboard which has been dried for 2 hours at 105° to 110° C. shall be weighed in a 400-ml beaker and 150 ml. of hot distilled water added The mixture shall be heated with frequent stirring, on a steam bath, until the fiberboard is completely wetted The mixture shall then be permitted to stand for 3 hours It shall be filtered through a Buchner type funnel (preferably sintered glass) and shall be washed five times with portions of distilled water large enough to cover the solid The filtrate shall be evaporated to a volume of approximately 150 ml., acidified with nitric acid and the chlorides precipitated by the addition of an excess of silver nitrate. After standing several hours in the dark, the precipitate shall be filtered into a weighed Gooch or glass crucible, washed with 0.01 N nitric acid, and dried at 110° to 150° C. The crucible shall be again weighed and the chlorides calculated as $ZnCl_2$.

4.3.9 *Ash*—A 2-gm. sample of small pieces of fiberboard shall be dried to constant weight at 100°–105° C and weighed to the nearest 1 mg The sample shall be ignited in a covered crucible of a material such as platinum, alundum, porcelain, or silica, which together with the cover has previously been ignited and weighed to 0.1 mg The ignition shall be accomplished at a low temperature, and the temperature shall gradually be raised to a maximum of about 925° C (1700° F) After this temperature is reached the lid of the crucible may be slid to one side until the combustion is complete When the fiberboard is completely burned the crucible shall be placed in a desiccator until its temperature is in equilibrium with that of the surrounding atmosphere. The crucible and its contents shall be weighed to the nearest 0.1 mg The ignition and weighing shall be repeated until the weight is constant.

4.3.10 *Flexibility*.—The test specimens shall be 6 inches long and 2 inches wide with the length dimension in the machine direction The specimens shall be bent 180° in the direction of the length over a mandrel 1 inch in diameter and examined for cracks and delaminations.

4.3.11 *Delamination*—Two sets of specimens of suitable size shall be prepared. One set shall be immersed for 24 hours in water at 70° to 75° F. The second set shall be baked for 24 hours in an oven maintained at a temperature of 212° ±2° F. Both sets shall be examined for delamination.

4.4 Lot tests for burning rate only.—When two successive lots of fiberboard supplied on a contract have been tested for all require-

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ments specified herein and found acceptable, succeeding lots may be tested only for burning rate and such check tests as the Government inspector deems desirable Fiberboard may also be accepted after testing only for burning rate if the contractor has consistently furnished acceptable fiberboard on one or more contracts or orders immediately preceding the lot in question, providing the contractor certifies that the fiberboard is identical in all respects to the acceptable material previously furnished. Such tests to determine burning rate shall be made at the place of manufacture in the presence and to the satisfaction of the Government inspector

4.5 **Rejection.**—If more than one of the three specimens tested fail to comply with a particular requirement the lot shall be rejected If one specimen fails, two additional specimens from the same lot sample shall be tested for the requirement in which failure was experienced Failure of either of the two additional specimens shall result in rejection of the lot

4.6 **Reinspection.**—A rejected lot may be submitted for reinspection after all deficiencies have been corrected by the contractor A lot submitted for reinspection shall be sampled as specified in 4.1.2 and tested for all requirements Two successive lots shall be so sampled and tested for all requirements If the two successive lots comply with all requirements additional lots may be accepted on the basis of tests for burning rate only.

5 PREPARATION FOR DELIVERY

5.1 **Packaging and packing.**—Unless otherwise specified, the material shall be packed in standard commercial containers of the type and kind commonly used for the purpose so constructed as to insure acceptance by common or other carriers for safe transportation, at the lowest rate, to the point of delivery.

5.2 **Marking.**—The containers shall be marked with the title and number of this specification, the name of the manufacturer, the contract or purchase order number and the weight. Shipments for the Navy shall be marked in accordance with the Navy Shipment Marking Handbook and for the Army in accordance with Specification 100–2.

6. NOTES

6.1 **Ordering data.**—Invitations for bids, and contracts or orders should specify the following:

a. Title, number, and date of this specification

b. Quantity (weight), size, and thickness of sheets

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto

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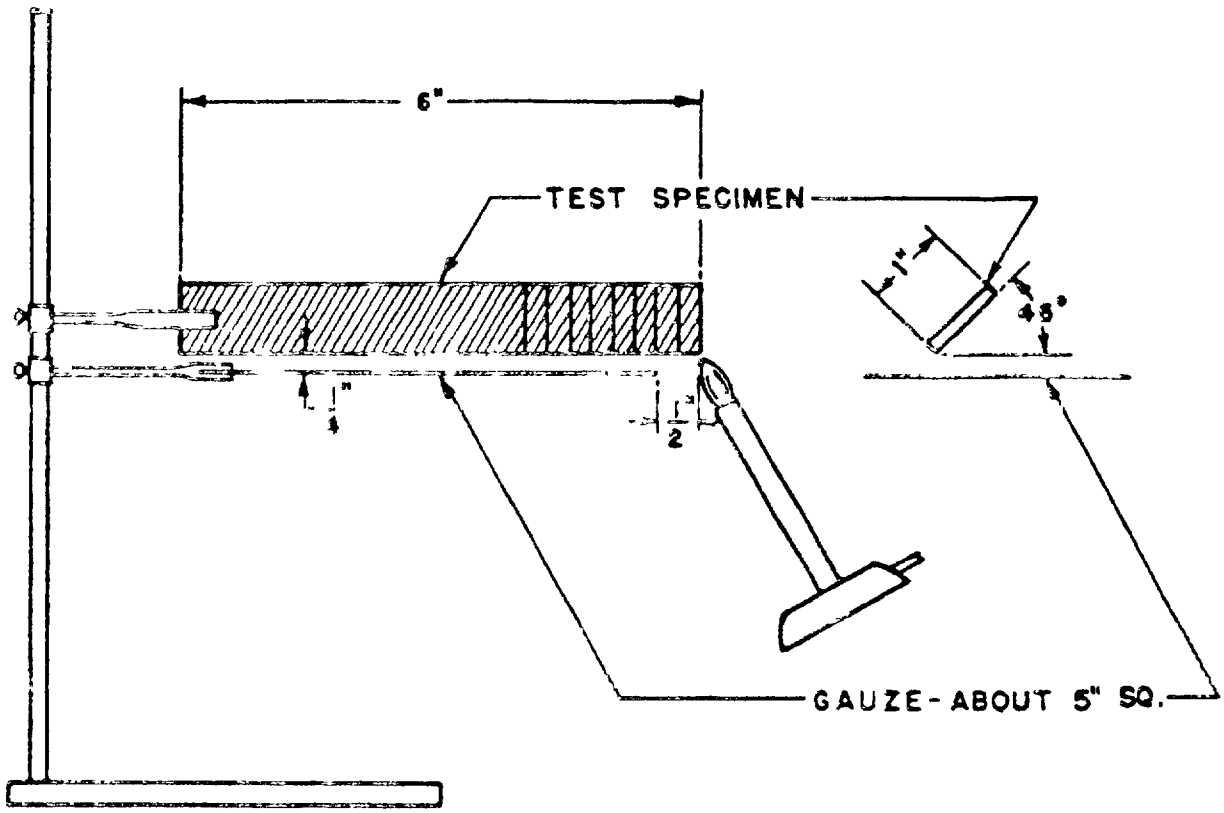
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NOTE:-
ALL DIMENSIONS IN INCHES. TOLERANCES ON FRACTION $\pm\frac{1}{64}$

FIGURE 1—*Apparatus for flammability test*

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(See Instructions Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City State ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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